1. Introduction

1.1. Objectives

The primary objective of the Recall App is to notify users when a product that they've logged in the app has been recalled. The secondary objective would be inclusion of best by/expiration notifications, to allow the user to know when their food has expired. The tertiary objective would be to integrate the feature to upload receipts to the system, allowing it to fill out pertinent information for the user. These objectives ensure that basic user needs are fulfilled by the system.

1.2. Scope

The scope of this document will encompass the entire Recall App. This will include functions that allow users bought items, updating FDA recalls, notifying when items are matched with recalls or when food expiration dates are approaching. These functions ensure that the application fulfills its objectives and is a useful resource for users. The final deliverable will be a cross-platform application.

1.3. Regulatory Concerns

The Recall App must adhere to industry regulations, including but not limited to:

- FDA Recall Guidelines (for food and drug recalls).
- CPSC Regulations (for consumer product recalls).
- GDPR and Data privacy laws for user data protection.

2. Program Requirements

2.1. Functions

Functions that our system will utilize:

- **User Account Management** – The system shall allow users to register, log in, and manage their preferences for notifications and tracking.
- **Manual Product Entry** – Users can input product details to check for recall status.
- **Purchase History Integration** – The system shall retrieve and store a user's past purchases and cross-reference them with active recalls.
- **FDA Recall Data Integration** – The system will use an API to pull recall data from the FDA's database.

- **Push Notifications** – The app will send automated alerts to users if a recall matches their purchase history or manually entered products.
- **Analytics Dashboard** – Users can view recall statistics, trends, and other analytics.
- **Computer Vision for Receipt Scanning (Future Enhancement)** – The app will use OCR (Optical Character Recognition) to extract product details from receipts.

2.2. Workflow

User Onboarding & Registration:

- Users create an account and set notification preferences.
    a. They can manually input purchase data or allow API-based history retrieval.

Recall Detection & Cross-Checking:

- The system fetches recall data from the FDA database.
- The system compares this data against the user's purchase history.
- If a match is found, the user receives an immediate notification.

User Interaction & Response:

- Users can acknowledge the recall notification.
- The app provides recommended actions (return, dispose, contact manufacturer).
- If applicable, users can coordinate donations for safe-to-consume recalled items.

Data Logging & Analytics:

- The system logs recall interactions for compliance tracking.
- Users and organizations can view recall trends through an analytics dashboard.

2.3. Performance & Scalability Requirements

To ensure the systems efficiency, the following performance criteria must be met:

Performance Standards:

- The app must support **at least 20 concurrent users** without noticeable system delays.
- Recall matching should occur within **2 seconds** of retrieving updated recall data.
- Notifications should be delivered within **5 seconds** of detecting a recalled product.

Scalability Considerations:

- **Efficient Data Storage:**
  - Product and recall data should be indexed in Firebase for **fast retrieval**.
- **Optimized Recall Matching:**
  - Future versions should include **batch processing** for managing large inventories.

Future Expansion:

- **Batch Scanning for Organizations:**
  - Community kitchens and food banks can scan multiple items at once.
- **Web-Based Dashboard:**
  - Expansion to a web interface for cross-platform accessibility.
- **Donation Coordination Feature:**
  - Users can mark food for donation before it expires.

3. Data Requirements

3.1. Type of Information

The system should be able to check a user's past purchase history and compare it to the FDA database. If a match is found between an item a user has purchased and an active recall, the system must notify the user of such a match.

Optional requirements included in the project is using Computer Vision to read receipts to automatically add the list of purchased items to the app. This would reduce the amount of time and effort that users would have to take to use the app, making the app easier to use.

3.2. Data Types and Formats

The system must store and process product information, whether entered manually by the user or retrieved via barcode scanning. The required attributes include:

- **Product Name** (string) – Name of the food item.
- **Brand Name** (string) – Manufacturer or brand of the product.
- **Category** (string) – Food type (e.g., dairy, meat, canned goods).
- **Expiration Date** (date) – When the item is expected to expire.
- **Lot/Batch Number** (string, optional) – Identifier used for recall matching.
- **Barcode (UPC/EAN)** (string) – Unique product identifier.

- **Date Added** (timestamp) – The date when the user added the product to inventory.
- **User ID** (string) – Links the product to a specific user.

3.3.1 Data Sources

- **Manual user input** – Users can enter product details directly.
- **Barcode scanning –** The app retrieves barcode data from an external product database.
- **Previously stored data** – The app remembers scanned items for faster future entry.

3.3.2 Recall Data

The system must retrieve, and process recalled information from the **FDA Recall API**.

Each recall entry must include:

- **Recall ID** (string) – Unique identifier for each recall event.
- **Product Name** (string) – Name of the recalled item.
- **Brand Name** (string) – Manufacturer responsible for the product.
- **Reason for Recall** (string) – Description of the issue (e.g., contamination, mislabeling).
- **Affected Lot/Batch Numbers** (list of strings) – Lot codes associated with the recall.
- **Recall Date** (date) – The official recall issue date.
- **Geographic Impact** (string) – Areas affected by the recall (e.g., nationwide, specific states).
- **FDA Case Number** (string) – Reference ID for FDA tracking.

3.3.2.1 Data Sources

- **FDA Recall API** – The app must periodically query the official FDA recall database.
- **User Reports** – Users may manually report additional recalls for verification.

3.3.3 User Data

Each registered user must have a profile storing relevant details:

- **User ID** (string) – Unique identifier for the user.
- **Email Address** (string) – Used for login and notifications.
- **Password (hashed)** (string) – Stored securely for authentication.
- **Notification Preferences** (Boolean) – Whether users want push notifications for recalls.
- **User Inventory** (list of product IDs) – Links stored food items to the user.

3.3.3.1 Data Sources

- **User Registration** – Users create an account manually.
- **OAuth Authentication** – Optional login via Google or Apple accounts.

3.3.4 System Logs & Analytics Data

The system must maintain logs for tracking user interactions, recall notifications, and app performance:

- **Log ID** (string) – Unique identifier for each system event.
- **Event Type** (string) – (e.g., product added, recall matched, user login).
- **Timestamp** (datetime) – When the event occurred.
- **User ID** (string) – Links the event to a specific user.
- **Response Time** (float) – How long the system took to process recall checks.

3.3.4.1 Data Sources

- **System Events** – Automatically generated logs from user actions.
- **Firebase Analytics** – Tracks app usage for performance monitoring.

Data Storage & Retrieval

4.1 Database Architecture

- The system must use **NoSQL Firebase Database** to store user inventories, recall matches, and notifications.
- Indexed searches must be implemented to **optimize recall matching speed**.
- Data must be structured to allow **real-time updates**.

4.2 Data Security & Privacy

- **User passwords must be hashed and stored securely**.
- **Access control mechanisms** must be in place to ensure that only authorized users can modify their data.
- Personal data should not be shared externally without **explicit user consent**.